



Piloting a data model for textile-to-textile recycling

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Part 1: Introduction

The textile industry has a significant environmental impact, contributing to global pollution through energy, water, and chemical usage, and accounting for 8-10% of global carbon emissions from apparel and footwear production¹.

Approximately 7 million kilograms of gross textile waste is generated within the EU and Switzerland on an annual basis². The disposal of this post-consumer textile waste in landfills or through incineration worsens the environmental challenge, highlighting the urgent need for effective solutions in end-of-life management strategies. The Textile Recycling Excellence, [T-REX](#) Project, supported by the Horizon Europe program, aims to guide the textiles industry in seizing business opportunities in closed-loop textile recycling using post-consumer textile waste feedstock.

¹ European Parliament (2022). The impact of textile production and waste on the environment. Available [here](#).

In November of 2023, the research team published the whitepaper titled [Connecting Threads: Assessing Digital Solutions and Needs for Circular Textiles](#), a first step in identifying, assessing and sharing key learnings about digital solutions, their types, needs and opportunities to support circular textile value chains. Adopting digital systems in the textiles industry offers benefits such as streamlined processes, enhanced transparency, collaboration facilitation, data-driven decision-making, and sustainability promotion. However, numerous challenges exist in transparency and standardisation of data structures and platforms, these must be addressed for effective implementation of sustainable tools. Moreover, the legislative environment in the European Union strongly urges the textile industry to adopt practices and uptake technologies to better support the creation of digital and physical infrastructure that support textile circularity.³

^{2,3} McKinsey (2022). Scaling textile recycling in Europe—turning waste into value. Available [here](#).

UNCOVERING INDUSTRY GAPS AND OPPORTUNITIES + PILOTING A SOLUTION

Through comprehensive primary and secondary research, three key areas were explored in the [whitepaper](#) – digital platform usage, data exchange protocols and challenges in the supply chain, and factors influencing the exchange of vital product data. This analysis yielded valuable insights into the opportunities and gaps associated with digitisation and circularity, with **nine overarching themes** identified.

The nine identified themes are highly interconnected and rely significantly on the industry's efforts to achieve data standardisation and system interoperability, which serve as crucial foundations for this digital transformation. In addition to these factors, various other enabling factors contribute to driving the industry's journey towards digitisation, such as legislation and compliance, certifications and standards, uptake and scaling of technology, and large-scale financial resources and upskilled human capital.

The research team set out to address these findings within the pilot, with the aim to standardise and digitise key data for chemical recycling, enabling more efficient textile-to-textile recycling. Textile waste information is largely shared via paper records distributed over email, with very little standardisation in data sourcing, formatting, or communication. Therefore, the team set out to develop and test an excel-based data model for sorters, preprocessing actors, and recyclers in the T-REX Consortium. Through doing so, the pilot specifically aimed to address the following themes identified in the whitepaper – Theme 1: Industry Alignment, Theme 2: Transparency Regarding Data, Theme 3: Trust Across Value Chain, Theme 7: Physical and Digital Infrastructure, and Theme 8: Building Capacity & Capabilities for Data and Digitation. By addressing these themes, the pilot aimed to lay foundations upon which further digitisation could be built.

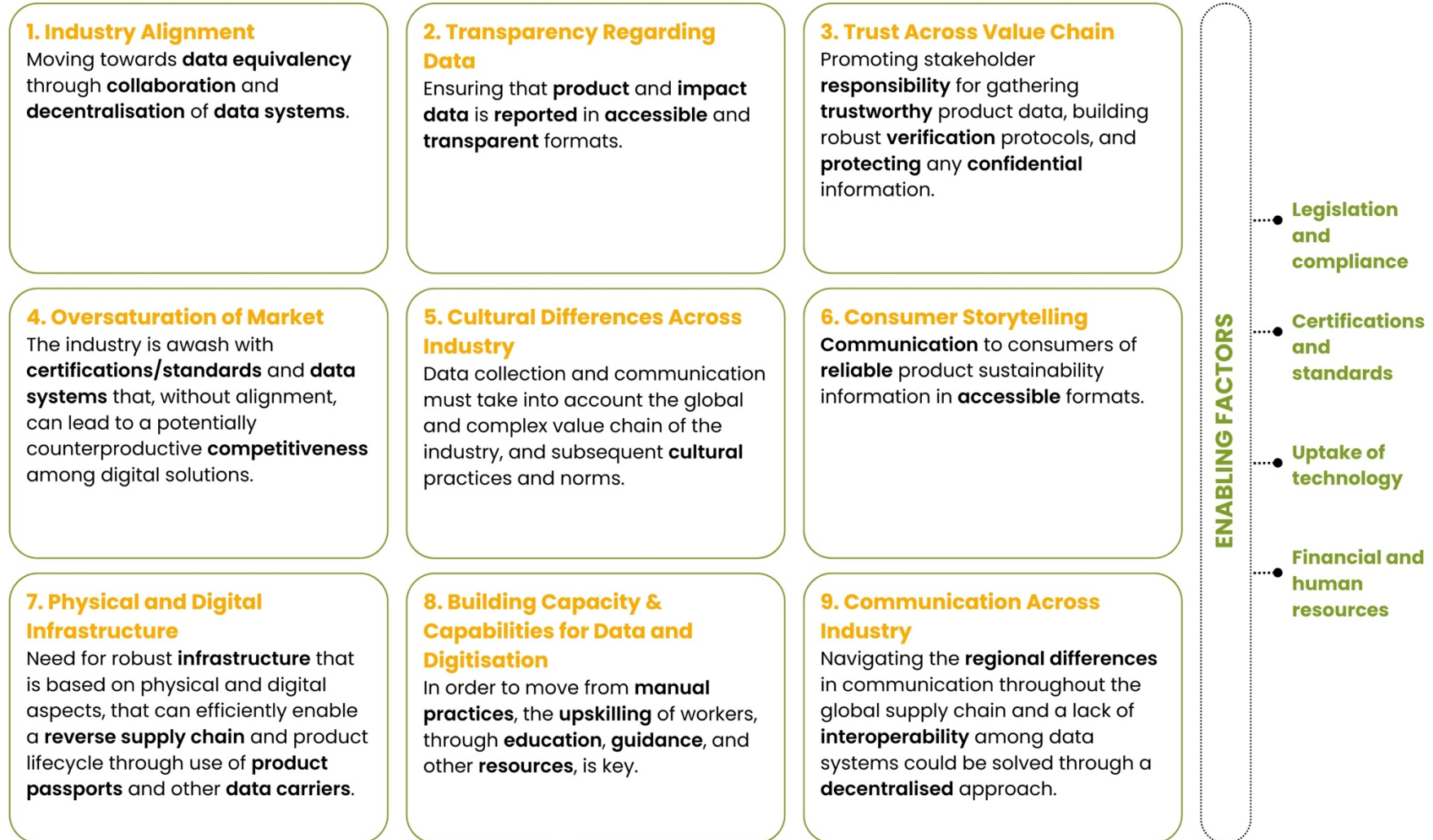


Figure 1: Nine overarching themes

Part 2: Pilot Summary

PILOT OBJECTIVES

Standardising data collection is an important aspect of the development of textile-to-textile recycling as it can help to structure the communication between economic partners in the textile recycling value chain. In the T-REX project, a blueprint was created for the textile-to-textile recycling of post-consumer household textiles. This provided the context for the pilot to research how data flows could trace the textile waste flowing through this recycling value chain.

The objective of the pilot was to **develop and test a data model** that would meet the **minimum data requirements** to capture the textile-to-textile recycling value chain within the T-REX project. The pilots began at the sorting stage and concluded at the recycling stage.

DEVELOPMENT OF THE DATA MODEL

Before designing the data model, the research team first investigated the minimum data requirements in the T-REX

waste value chain (sorting, preprocessing, and recycling). As part of this foundational research, an initial list of data points was developed based on existing Digital Product Passport (DPP) data protocols from TrusTrace, circular.fashion and EON.

Building on these protocols, the research team conducted several interviews with consortium members and external partners to identify which data points they independently collect, exchange and consider the highest value to their chemical recycling process. This shaped the information that was captured with the minimum data requirements of the data model.

Ensuring the data model's digital interoperability was a crucial next step in its development. TEXroad, a digital solutions organisation specialising in data-driven improvements to textile management infrastructure, provided expert guidance to the research team on implementing these measures. The table below details the findings from this development phase.

Sorting		
Data Field Name	Pilot Findings	Further Research
Showstopper	A showstopper is a material that renders an entire batch of textile waste unusable for recycling when present. The idea of incorporating showstopper data into the model was explored to encourage sorting facilities to check for these materials in each batch. Recyclers also saw it as a valuable data point. However, it was ultimately excluded due to findings that sorting facilities currently lack the technical and economic means to provide this level of accuracy.	The material complexity of textile waste makes the identification of showstoppers difficult. A Digital Product Passport (DPP) could potentially be a valuable resource to identify all materials present in a garment, allowing showstoppers to be more easily identified and removed during sorting.
Hazardous Materials	For traceability and safety purposes, recyclers would like to know if there are hazardous materials present in their recycling feedstock. Similarly to showstoppers, sorters lack the means to accurately identify and sort by hazardous materials. Therefore, this data point was also excluded from the data model.	See above.
Recycling		
Data Field Name	Pilot Findings	Further Research
–	It's a common practise on an industrial scale for recyclers to mix feedstock from different batches to achieve the desired feedstock quality. While the data model is equipped to capture this complexity, it would require a digital system to follow and combine the batches, as well as connect them to the material flows in the real world. This level of complexity was however not within the scope of the T-REX project.	More research should be conducted on how similar data from different data systems can be aligned and exchanged so data flows in the digital world match material flows in the real world.

Table 1: Findings of the data model development phase

THE T-REX DATA MODEL

The data model illustrates the minimum data requirements within the T-REX textile recycling value chain. As a result, the data model should not be seen as a final end-product. Instead, it should serve as a living document for stakeholders in the textile waste supply chain to build upon for data standardisation research.

The complete T-REX data model can be found in [Annex I](#).

It consists of the following four groups of data:

- **Entity Data** – data points describing the economic entity that is processing materials
 - o Example: company name, address and registry information
- **Batch Data** – data points that identify and quantify a distinct batch of materials as it is processed by a specific entity and/or it flows along the value chain
 - o Example: the batch ID number, weight, activities carried out during processing steps, types of downstream stakeholders in the value chain that can receive the outputs, and the quantity of each output batch based on these types of downstream stakeholders
- **Output Specifications** – data points that characterise batches of materials and/or define the material requirements of a step in the value chain
 - o Examples of sorting output specifications: sensor accuracy, feedstock material composition
 - o Examples of pre-processing output specifications: feedstock format (full garments, cut pieces, fibres, pellets, etc.)
 - o Examples of recycling output specifications: recycled content in output, feedstock recyclability, and technical documents on output material (Certificate of Analysis; Material Safety Data Sheet; Technical Data Sheet).
- **Material Streams** – data points that are used in place of a data system that connects the value chain, these are data points that describe the streams of textile waste arriving at the collector
 - o Example: total collection quantity, from which countries it came and which of that was internally collected or externally purchased.

TESTING OF THE DATA MODEL

To test the data model, several interviews were conducted with consortium members and external partners, to cover each step in the waste supply chain. The T-REX project focuses on three material streams: cotton (recycled by Infinited Fiber Company), polyester (recycled by CuRe Technology), and polyamide 6 (recycled by BASF). For each of these streams, one of the performed pilot runs was captured with the data model, of which the results are visualised in the flow charts below.

It should be noted that the data visualised in these flow charts is based on pilot scale processes that, within the T-REX project, also ran in an experimental setting with small batch quantities. As a result, the data (especially the Batch & End Point Weights) are only indicative of the T-REX project and cannot be taken as representative of the operations of the entities.

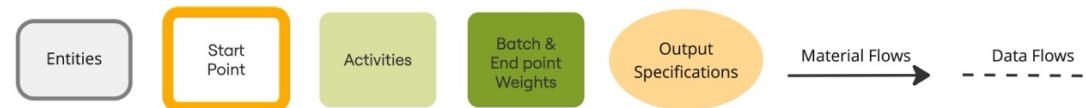


Figure 2: Legend

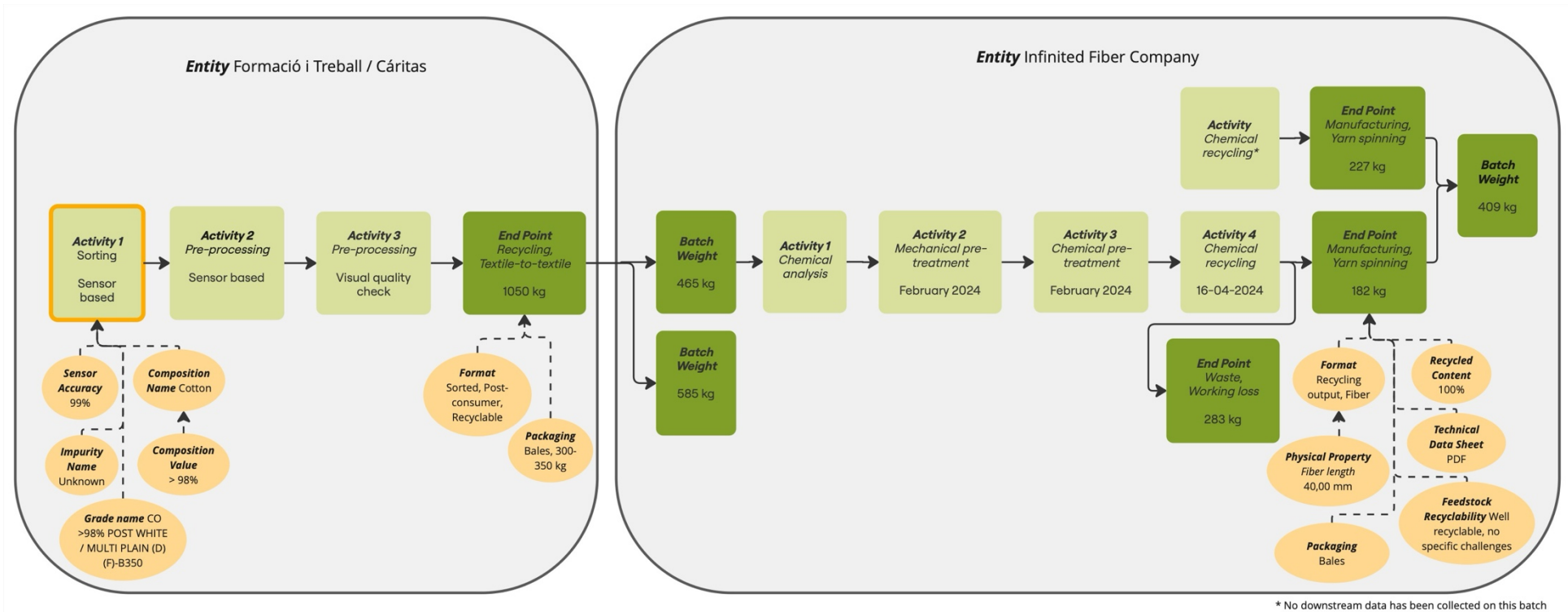


Figure 3: Cotton recycling data stream

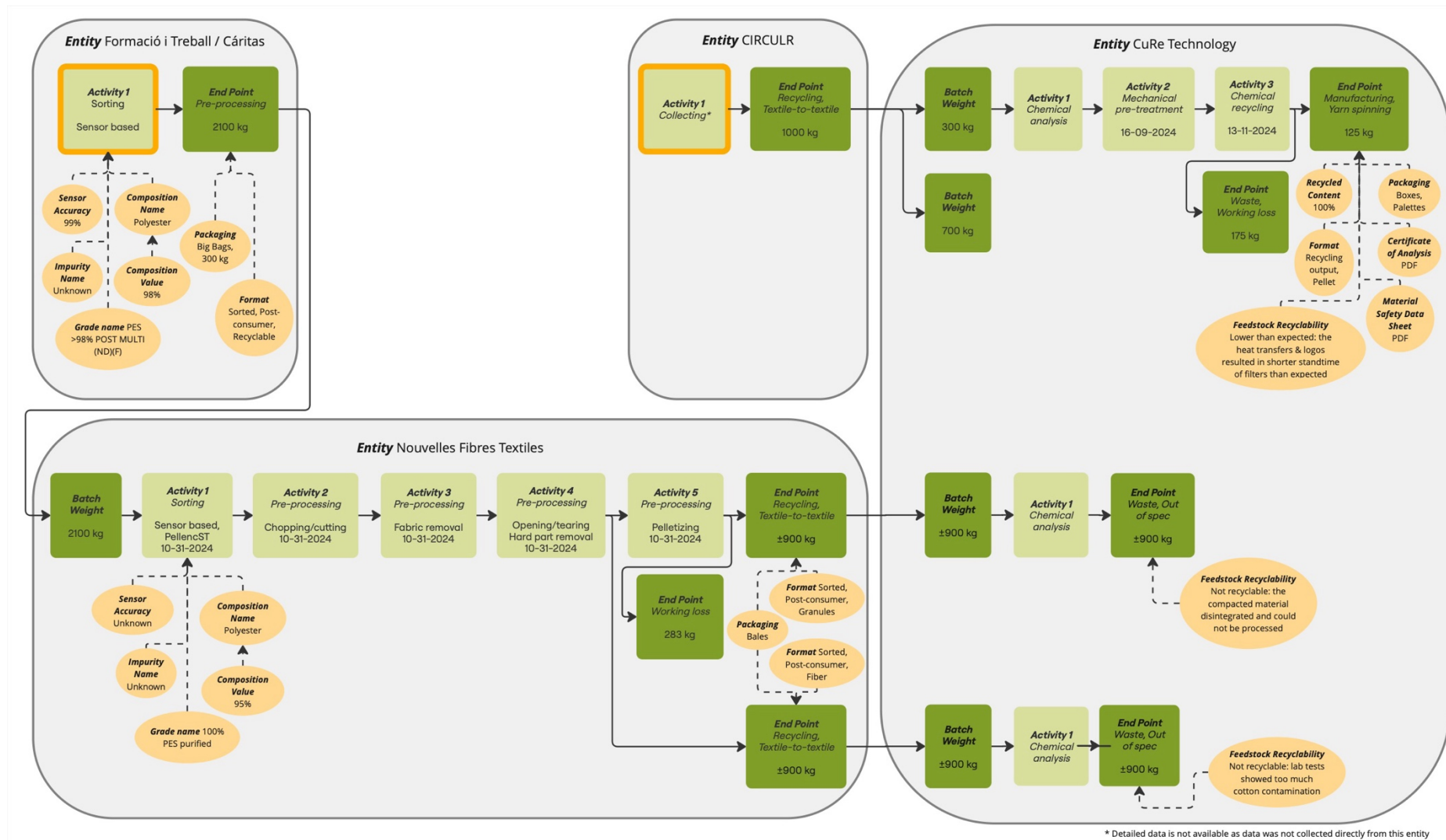


Figure 4: Polyester recycling data stream

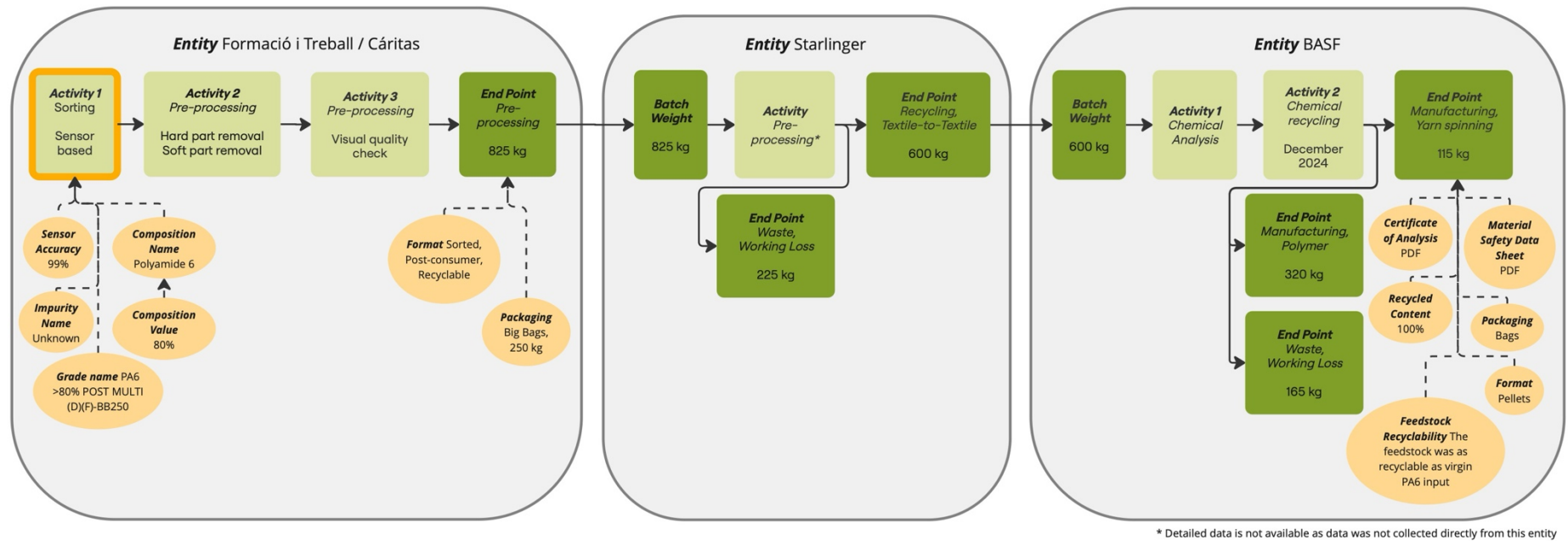


Figure 5: Polyamide 6 recycling data stream

The goal of the data collection portion of the pilot was to (1) investigate if it was possible to receive critical datapoints (2) determine how to structure this data, (3) receive feedback on specific data points within the data model.

Each stakeholder interview consisted of collecting data through use of the data model and receiving feedback on the data model. See below for the findings from this testing phase:

Collecting			
Data Point ID	Data Field Name	Pilot Findings	Further Research
400,00-412,XX	Material Stream	Collection was not within the scope of the pilot. Nonetheless, data points were developed to document the collection quantities and origins of feedstock in the project. The collectors that were interviewed were able to provide the data requested.	Development of the data model to document collection data so that subsequent data systems can digitally trace existing material flows
Sorting			
Data Point ID	Data Point ID	Pilot Findings	Further Research
201,10	Activity Date	Since sorting is often a continuous process, sorters cannot always provide specific dates for activities. This is especially the case when they work with an existing sorting category.	Define how value chain actors should document processing of materials over time to increase the quality and reliability of the available data.
310,00	Sensor Accuracy	Sensor accuracy may be considered confidential information by the company that manufactured the automated sorting technology.	

321,10	Composition Value	Sorters use NIR scanner settings to define the Composition Value. They are unable to provide a more accurate measurement of fibre composition, however they noted that feedstock requirements were typically met under such settings.	Document value chain expectations for composition data of sensor-sorted batches, and how to include in data model.
330,00–330,20	Contamination	Sorters were unable to provide details on feedstock contamination in the sorted batch. Some NIR scanning technology can detect this on a garment level, but this information is not recorded at a bale level.	See above.
Pre-processing			
Data Point ID	Data Field Name	Pilot Findings	Further Research
201,00	Activity	Within the current structure of the data model, activity values can only be filled in as separate data points (Activity 1, Activity 2, etc.) but often activities will happen simultaneously. As a result, the data flows do not give a completely accurate description of the material flows.	Development of activity values list to allow for a multiple selection format.
Recycling			
Data Point ID	Data Field Name	Pilot Findings	Further Research
201,00	Activity	This is difficult to standardise as recyclers employ a variety of approaches to recycle textile feedstock, often changing with scale.	Test the representativeness and granularity requirements of the activity values list for recycling.

201,10	Activity Date	Recyclers typically only communicate the production date, not that of each step of the process.	Assessment as to if this level of data is necessary.
211,00	End Point (Out of Spec)	The data model defined “End points”, of which “Waste” is one, which again has several sub-values: “Out of Spec” and “Working Loss”. Recyclers required clarity on how to define these terms.	Standardize “Out of Spec” and “Working Loss” on a recycling level.
	End Point (Working Loss)	Recyclers can be hesitant to share the working loss (e.g. yield) as it can be considered sensitive information.	
360,00	Recycled Content	Recyclers required clarity as to whether recycled content percentage in the output batch included additives, or just the target polymer.	Develop data points that could trace the non-recycled materials added to the recycled batch.
370,00– 372,00	Certificate of Analysis (COA); Material Safety Data Sheet (MSDS); Technical Data Sheet (TDS)	Not all recyclers provided these documents.	Better integrate uploaded documents (COA, MSDS and TDS) into to the data model.

General Pilot Findings			
Data Point ID	Data Field Name	Pilot Findings	Further Research
-	-	Stakeholders in the textile waste supply chain (aggregators, sorters, preprocessors, and recyclers) often perform a variety of intersecting activities	Reflect the overlapping activities of waste supply chain actors in the data model.
200,10	Batch Weight	It's important to clarify whether this refers to the net weight, defined as the weight of the feedstock without packaging.	Establish a standard system for reporting batch weight

Table 2: Findings of the data model testing phase

Part 3: DPPs in the Textile Waste Supply Chain

The Digital Product Passport (DPP) has the potential to provide key data points for the textile waste supply chain, but this requires a strong data communication foundation into which DPPs can be integrated.

DPPs are specifically helpful for providing sorters with product information to inform sorting decisions. Near Infrared (NIR) scanners are currently used in the sorting process to determine the composition of the main fabric of a garment, but have limitations in accuracy and trim identification. DPPs can offer a solution to this by providing verified information on material composition, finishes, prints,

and trims which can help sorters better grade feedstock per recycler requirements.

However, a hurdle of DPP data is that it is connected to individual garments, whereas textile waste is gathered on a batch or shipment level. Existing protocols result in garments arriving in batches to sorting centres, making it hard for sorters to gather individual garment-level datapoints within a textile waste batch and communicate critical data to downstream stakeholders. During subsequent preprocessing and recycling steps, textile feedstock is physically modified thus removing the DPP entirely until a new product is made requiring a DPP of its own.

Part 4: Next Steps

The aim of this pilot was to digitise the **T-REX waste supply chain** with the goal of capturing and somewhat standardising the critical data that flows through the sorters, aggregators, and preprocessors, through to the recyclers. The pilot testing of the data model, therefore, took place in an **experimental** setting with this small group of supply chain stakeholders and under specific assumptions.

Furthermore, a data system is required to move data documented by the developed data model across the value chain by aligning the data flows in the digital world with the material flows in the real world. While the data model structures and holds the information, it is the data systems that allow the data to move across the value chain. As such, this textile waste data model and its subsequent data system would have to be **further optimised to be used on an industrial scale**, a scale that realistically contains multiple intersecting waste flows, a network of supply chain stakeholders, and many data communication intricacies.

In addition to the specific next steps given in Part 2 of the report, the following complexities were not explored within this data model and merit further research:

- Applying the data model in the supply chain by **following a specific flow of textile waste** from aggregation, sorting, all preprocessing steps, and recycling. While the ambition of the pilot was to execute this within the T-REX project supply chain, the **intricacy** of the feedstock flows and **lack of data systems** made it hard to track a specific batch.
- Building the associated data system to **extend past batch level**. The T-REX project focused on specific feedstock batches per recycler but in practice and especially as recycling scales, recyclers will combine and store feedstock batches from many sources on a **shipment level**. The data model should be able to follow the movement of feedstock through these complexities and realities of the textile waste supply chain.
- Ensuring the data from the reverse supply chain is **interoperable** with data models and digital systems currently in use in the **product creation process** and existing supply chain. This will allow for relevant recycled content material data to be communicated through the fibre, yarn, fabric, and garment manufacturing steps.

- Ensuring interoperability between **traceability platforms** and **certification schemes** by aligning data models.
- Denoting **feedstock sources** (Post-Industrial, Pre-Consumer, and Post-Consumer) that are typically used in textile-to-textile recycling. The T-REX project was focused solely on post-consumer waste, but textile recyclers typically will use a combination of these three source types for operations. Non-textile feedstocks (such as bottles, packaging, etc) can also be included as it is somewhat common for recyclers in the value chain to combine their feedstock sources.

Iterative work in the space of digitisation is necessary to establish a strong foundation of data sharing in the textile industry, especially data around textile waste and recycled content. Projects such as EU Horizon's [PESCO-UP](#) aim to further build digital capabilities for textile waste supply chain data by developing tools and practices for data sharing between relevant supply chain actors.

ACKNOWLEDGEMENTS

The research team would like to thank Traci Kinden of [TEXroad](#) for sharing her expertise and providing guidance during the planning and execution of this pilot and subsequent report. Thank you also to the T-REX consortium and external project partners for their participation in the pilot testing, namely the teams from ModaRe Caritas, Valvan, Nouvelles Fibres Textiles, Veolia, CuRe Technology, BASF, and Infinited Fiber Company.

Annex 1: T-REX Data Model

Entity Data					
Data Category	T-REX_ID	T-REX Data Field Name	Examples (not a complete values list)	Data Format	Definition
Entity Details	100,10	Entity Name	Trident Courage	Open text (brief)	Registered name of company being described. 100 series is generic "Entity" data field to be used with any company type
Entity Details	100,21	Street	Bankova Str.	Open text (brief)	Street, house number, city, postal code, and country must appear together.
Entity Details	100,22	City	Kyiv	Open text (brief)	Street, house number, city, postal code, and country must appear together.
Entity Details	100,23	House Number	11	Open text (brief)	Street, house number, city, postal code, and country must appear together.
Entity Details	100,24	Province / Region	Kyiv	Open text (brief)	Street, house number, city, postal code, and country must appear together.
Entity Details	100,25	Postal Code	01220	Open text (brief)	Street, house number, city, postal code, and country must appear together.
Entity Details	100,26	Country	Ukraine	Text (from standard list)	Street, house number, city, postal code, and country must appear together.
Entity Details	100,30	Registry name	GLN OSH Dutch Chamber of Commerce	Text (from standard list)	The name of the registry used to identify the company or facility listed (e.g. OSH, GLN, Dutch Chamber of Commerce, EU-VAT, etc).
Entity Details	100,40	Registry Other		Open text (brief)	Used when registry name is not available from standard list.

Entity Details	100,50	Registry ID Number	FR202206770RY8G	Alphanumeric text (fixed format)	The number in the registry named in 100,3 or 100,4 which is used to identify the company or facility listed.
Entity Details	100,60	Entity Type	Collector Sorter Pre-processor Recycler	Text (from standard list)	Determined by activities carried out.

Batch Data					
Data Category	T-REX_ID	T-REX Data Field Name	Examples (not a complete values list)	Data Format	Definition
Batch Details	200,00	Batch Number	123456	Alphanumeric text	Number assigned by the data manager or by the materials owners to track batches of materials at any level. This could be a small bale (55kg), a container, etc. A PO number may also be used.
Flow and Batch Details	200,10	Batch Weight	10	Number (fixed format)	The weight of the batch being described in available unit.
Flow and Batch Details	200,20	Batch Weight Unit	tonnes	Text (from standard list)	Unit of measurement used for the Batch Weight
Flow and Batch Details	201,00	Activity 1	Sorting, Sensor based, TrinamiX	Text (from standard list)	Defines what activities took place with a specific batch of materials.
Flow and Batch Details	201,10	Activity 1 Date	15-apr-24	Date format (standard)	Date a specific activity took place. This may be a specific day, or data may be limited to a month or a year
Flow and Batch Details	202,XX	Activity 2...			

Flow and Batch Details	210,00	Packaging	Bales Boxes Capsacks	Text (from standard list)	How the batch or shipment is packed
Flow and Batch Details	211,00	End Point 1	Next step, Fine sort	Text (from standard list)	Defines where the batch of materials goes. This can be expressed as a single data field (as seen here) or broken down into several.
Flow and Batch Details	211,10	End Point 1 Quantity	5000	Number (fixed format)	Numeric value for the amount of materials that is included in End Point 1 (211,00)
Flow and Batch Details	211,20	End Point 1 Quantity Unit	kg Tonnes	Text (from standard list)	Unit of the quantity in End Point 1 Quantity (211,10)
Flow and Batch Details	211,20	End Point Packaging	Includes Excludes	Text (from standard list)	Defines if the End Point Quantity (211,10) includes or excludes packaging weight
Flow and Batch Details	212,XX	End Point 2...	Recycle, Textile to textile	Text (from standard list)	Multiple end points are possible for a single batch.

Output Specification Data					
Data Category	T-REX_ID	T-REX Data Field Name	Examples (not a complete values list)	Data Format	Definition
Input Output Specifications	300,00	Grade Name	IFC Class 1 White Cotton 95% Super Denim Blue	Alphanumeric text	Written name, determined by the actor producing the outputs or the actor buying the inputs
Input Output Specifications	310,00	Sensor Accuracy	95%	Number as a % (fixed format)	Defines the accuracy level and / or requirements of materials batches or of equipment. In the case of a sensor, this is accuracy as provided by the manufacturer.

Input Output Specifications	311,00	Sensor Accuracy Method	Fiber composition from representative sample size of sorted textile batches was lab tested using a dissolution method	Open text (multi-line)	Description of the methodology used to calculate the accuracy level. In the case of supporting documents, a link to the data repository where they are stored can be provided.
Input Output Specifications	320,00	Format	Recycling feedstock, post-consumer, fiber	Text (from standard list)	Describes the source and construction or format of the materials
Input Output Specifications	321,00	Composition Name	Cotton	Text (from standard list)	Name of the primary textile fiber or other material compositions (e.g. name of the monomer) according to a standardized list
Input Output Specifications	321,10	Composition Value	96,50%	Number as a % (fixed format)	% of the total by weight for Composition Name (321,00)
Input Output Specifications	321,20	Composition Threshold	Minimum Maximum Exact	Text (from standard list)	For input specifications Minimum, maximum, or exact requirements for composition
Input Output Specifications	330,00	Contamination 1 Name	Animal fibers Polyurethane Elastane	Text (from standard list)	For both input and output specifications The name of the contaminant; this selection dictates further information requirements such as amount and unit Connected to a specific batch number (200,00). If more than one contaminant is defined, each

					should be numbered sequentially (e.g. 331,00; 332,00; 333,00; etc.)
Input Output Specifications	330,10	Contamination 1 Amount	5,00	Number as a % (fixed format)	For both input and output specifications Numeric quantification or text description of contamination level (how much is present)
Input Output Specifications	330,20	Contamination 1 Threshold	Minimum Maximum Exact	Text (from standard list)	Minimum, maximum, or exact requirements for composition
Input Output Specifications	331,XX	Contamination 2 Name...			
Input Output Specifications	340,00	Physical Property 1 Name	Fiber length Piece size	Text (from standard list)	For input and output specifications Describes physical properties that are quantified in additional data fields
Input Output Specifications	340,10	Physical Property 1 Measure	5	Number (fixed format)	Numerical value used to measure physical property
Input Output Specifications	340,20	Physical Property 1 Unit	mm	Text (from standard list)	Unit of the numerical value measurement
Input Output Specifications	340,30	Physical Property Verification Method	Microscopic evaluation	Open text (multi-line)	Equipment or testing method used to measure physical property
Input Output Specifications	341,XX	Physical Property 2...			

Input Output Specifications	350,00	Feedstock Recyclability	Well recyclable, no specific challenges	Open text (multi-line)	Description of relative recyclability of feedstock batch as determined by recycler
Input Output Specifications	360,00	Recycled Content	100%	Number as a % (fixed format)	Describes percentage of recycled content of target polymer (e.g. PET, PA, cellulose, etc) in final output. Does not include comonly added chemistries or additives.
Input Output Specifications	370,00	Certificate of Analysis (COA)		PDF	Document provided by the recycler that verifies specifications, quality, and compliance of their output material based on test results from laboratory analysis.
Input Output Specifications	371,00	Material Safety Data Sheet (MSDS)		PDF	Document provided by the recycler that provides information on the properties, hazards, safe handling, storage, and emergency measures for their output material.
Input Output Specifications	372,00	Technical Data Sheet (TDS)		PDF	Document provided by the recycler that gives detailed information on the properties, performance characteristics, applications of the output material.

Material Streams Data					
Data Category	T-REX_ID	T-REX Data Field Name	Examples (not a complete values list)	Data Format	Definition
Material Streams	400,00	Total Collection Quantity	900	Number (fixed format)	Numeric value for the total collected post-consumer textile waste that came into the sorting facility in the determined 6 months time frame
Material Streams	400,10	Total Collection Quantity Unit	tonnes	Text (from standard list)	Unit of the quantity in Total Collection Quantity (400,00)
Material Streams	401,00	External Purchase Quantity	400	Number (fixed format)	Numeric value for the post-consumer textile waste that came into the sorting facility in the determined 6 months' time frame and that was purchased from external collectors
Material Streams	401,10	External Purchase Quantity Unit	tonnes	Text (from standard list)	Unit of the quantity in External Purchase Quantity (401,00)
Material Streams	402,00	External Purchase Country 1 Name	Germany	Text (from standard list)	Name of the country where the external collector is located

Material Streams	402,10	External Purchase Country 1 Quantity	30	Number (fixed format)	Numeric value for the post-consumer textile waste that was purchased from the collector based in the country determined in External Purchase Country 1 Name
Material Streams	402,20	External Collection Country 1 Quantity Unit	%	Text (from standard list)	Unit of the quantity in External Collection Country 1 Name
Material Streams	403,XX	External Collection Country 2 Name...			
Material Streams	410,00	Internal Collection Quantity	500	Number (fixed format)	Numeric value for the post-consumer textile waste that came into the sorting facility in the determined 6 months' time frame that was collected through the company's own collection bins and additional internal collection systems
Material Streams	410,10	Internal Collection Quantity Unit	tonnes	Text (from standard list)	Unit of the quantity in External Collection
Material Streams	411,00	Internal Collection Country 1 Name	Germany	Text (from standard list)	Name of the country from where the internal collection bins are located

Material Streams	411,10	Internal Collection Country 1 Quantity	80	Number (fixed format)	Numeric value for the post-consumer textile waste that was collected through the company's own collection bins and additional internal collection systems in the country determined in Internal Collection Country 1 Name
Material Streams	411,20	Internal Collection Country 1 Quantity Unit	%	Text (from standard list)	Unit of the quantity in External Collection Country 1 Name
Material Streams	412,XX	Internal Collection Country 2 Name...			